ESS SENTINEL 2.0

Evaluation report

2026-09-29T09:25:30.533313+00:00

Source: Synthetic reference dataset

Batch: batch-732c43b47f014f6e

Model: ESS-2026-09-29T050332-ebf09b1e

Screening stage: 24h

Decision-support prototype. Synthetic results are not real-world qualification evidence. Final disposition requires qualified engineering review.

Regression comparison

parameter	model	split	MAE	RMSE	R2	MedianAE
Leakage current	Linear	validation	4.2536	8.9203	0.55329	2.1322
Leakage current	Ridge-1	validation	4.2825	8.9528	0.55003	2.1261
Leakage current	Ridge-10	validation	4.2848	8.9573	0.54957	2.1292
Leakage current	RandomForest	validation	3.0437	8.315	0.61186	0.66957
Leakage current	GradientBoosting-2	validation	1.9488	8.1242	0.62946	0.18107
Leakage current	GradientBoosting-3	validation	1.9086	8.0687	0.63451	0.1632
Leakage current	Linear	test	3.7531	8.3993	0.56329	1.9476
Leakage current	GradientBoosting-3	test	1.6839	7.814	0.62203	0.15807
Propagation delay	Linear	validation	3.4213	7.7264	0.52266	1.7391
Propagation delay	Ridge-1	validation	3.4618	7.7417	0.52078	1.7504
Propagation delay	Ridge-10	validation	3.4618	7.7502	0.51972	1.7322
Propagation delay	RandomForest	validation	2.4118	7.0325	0.60455	0.65764
Propagation delay	GradientBoosting-2	validation	1.5264	6.9387	0.61503	0.11955
Propagation delay	GradientBoosting-3	validation	1.5425	6.9513	0.61363	0.11744
Propagation delay	Linear	test	3.0419	6.9021	0.55174	1.5276

parameter	model	split	MAE	RMSE	R2	MedianAE
Propagation delay	GradientBoosting-2	test	1.3361	6.1109	0.64861	0.12889
Standby current	Linear	validation	0.86804	1.8786	0.54248	0.44166
Standby current	Ridge-1	validation	0.88383	1.876	0.54376	0.46018
Standby current	Ridge-10	validation	0.88235	1.873	0.54517	0.45764
Standby current	RandomForest	validation	0.59625	1.7061	0.62262	0.15432
Standby current	GradientBoosting-2	validation	0.40479	1.7139	0.61918	0.032478
Standby current	GradientBoosting-3	validation	0.40331	1.7199	0.61652	0.032662
Standby current	Linear	test	0.73556	1.7622	0.54559	0.32431
Standby current	GradientBoosting-3	test	0.33846	1.5925	0.62892	0.029016

Methodology and limitations

Robust peer statistics, early-feature Isolation Forest and lot-disjoint regression support transparent review rules. Scores are not failure probabilities. Calibration intervals are approximate under lot dependence. Synthetic evidence is not production validation. Review states do not alter engineering measurements or decisions.